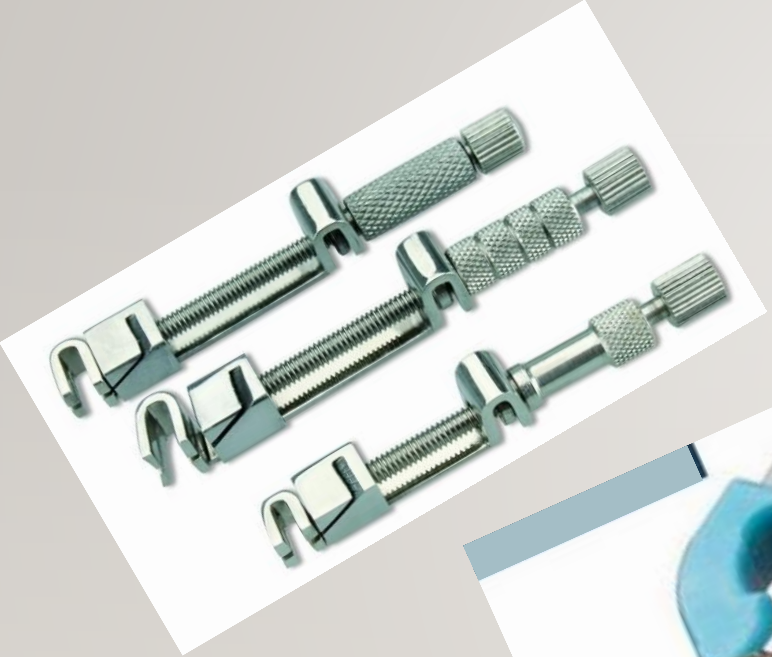
Dental matrices and wedges

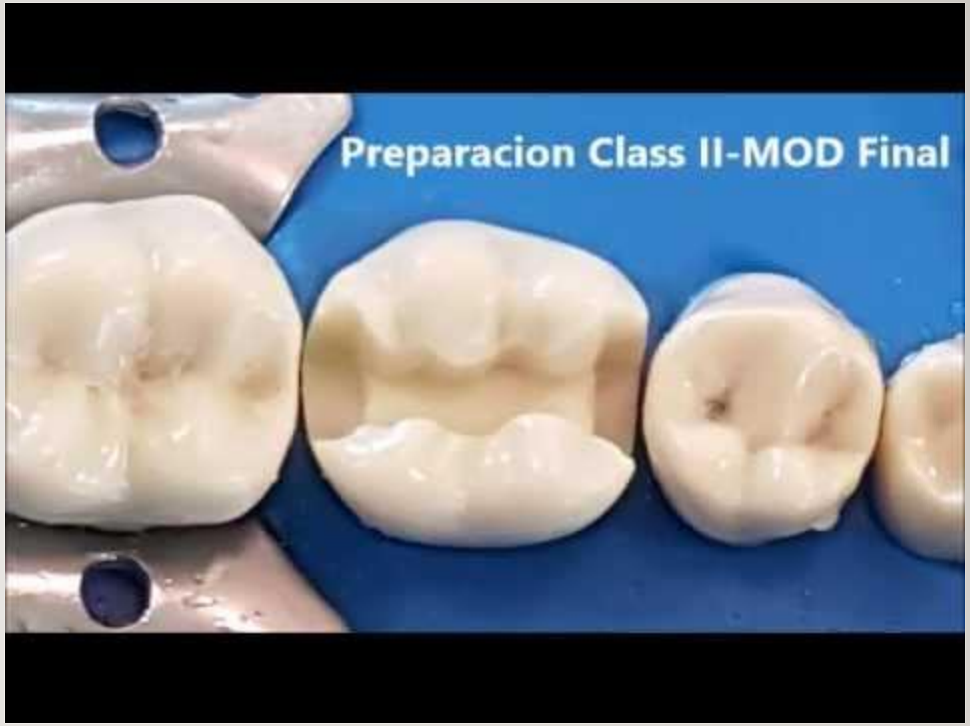


ILO's:

By the end of the lecture the student should be able to:

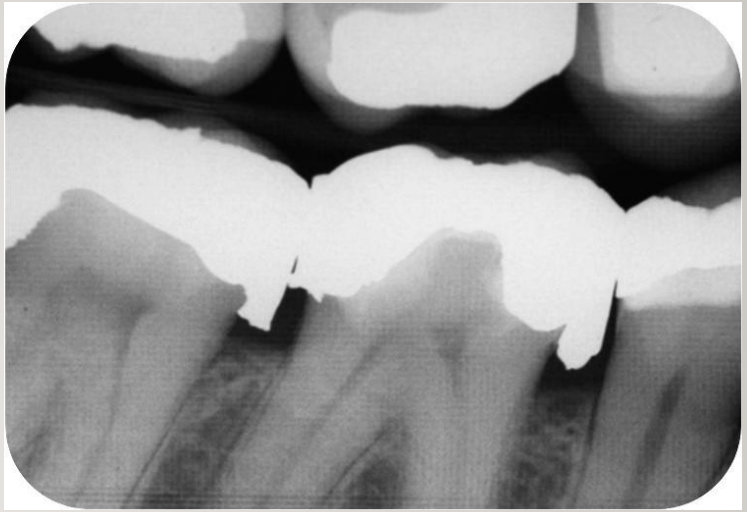
Describe about Objectives and ideal requirements of matricing
Explain the various forms of matricing
Describe the purposes and type of wedges
Explain about various wedging techniques

Class- II

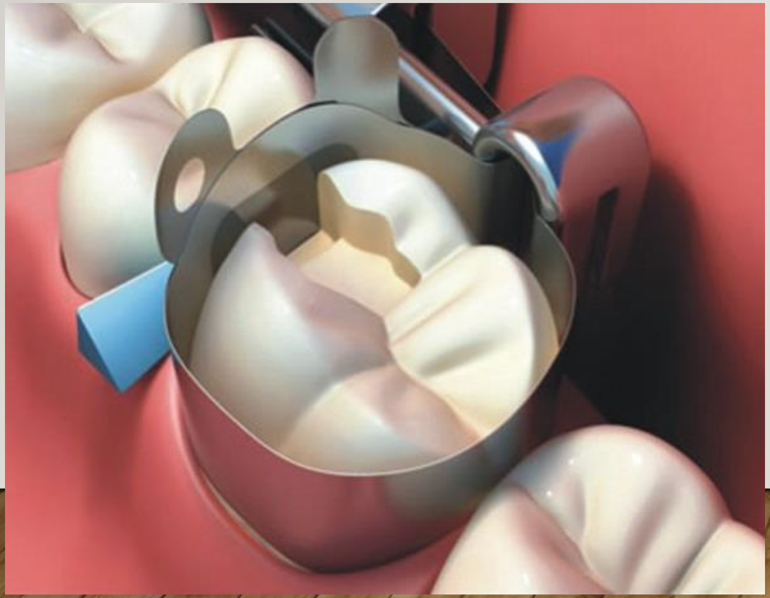


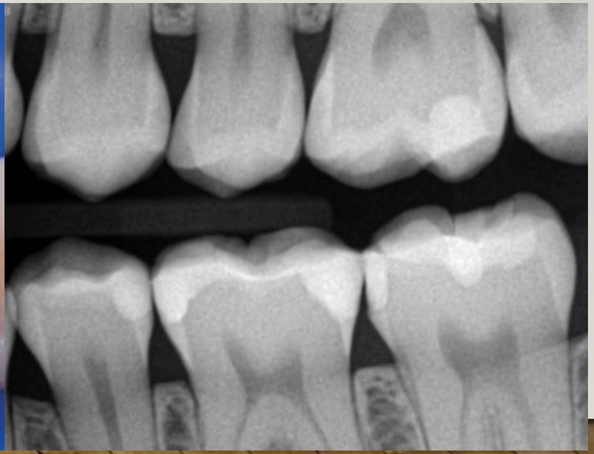
Class- III or IV

Not using Matrix & Wedge

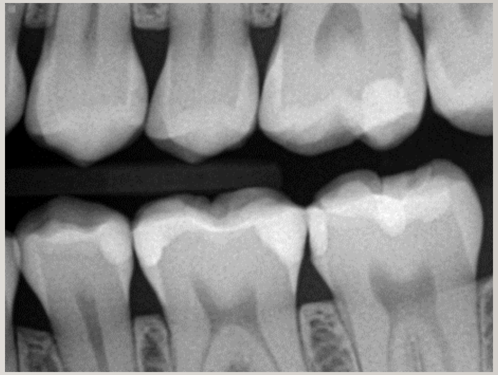


Using Matrix & Wedge





Why we need Proper Contacts and Contours



- Prevents the food from impinging between the teeth
- Prevent periodontal problems as well as formation of dental caries
- Stabilizes the dental arches
- Distributes the forces of mastication.

What is matricing?

Matricing is a procedure wherein a temporary wall is created opposite to axial walls, surrounding areas of tooth structure that were lost during preparation

The primary objective of the matrix is to restore anatomic contours and contact areas.

The Ideal requirements of a matrix include :

(1) Rigidity

(2)Versatility (The matrix band should be able to adapt to almost any size and shape of tooth)

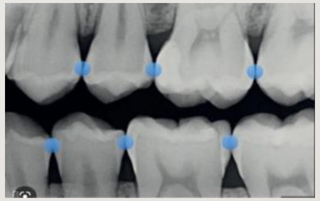
(3) Establishment of proper anatomic contour

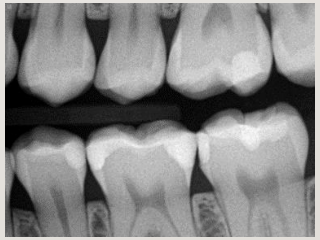
(4) Restoration of correct proximal contact

(5) Prevention of gingival excess(Overhanging)

(6) Ease of application

(7) Ease of removal





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MATRIX SYSTEMS

1. Universal Matrix (Tofflemire Matrix)

2. Pre-contoured matrix(Sectional matrix)

3. Auto-matrix

4. Celluloid / Polyester strips

UNIVERSAL MATRIX (TOFFLEMIRE)

BY, B.R.TOFFLEMIRE

✓**Commonly used to restore Class II preparations.**

2 parts

Matrix Retainer - It is an instrument used to hold matrix band in position

Matrix band

✓**Disadvantage - Flat band must be contoured.**

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https://www.youtube.com/watch?v=upzeWfgvpQ8

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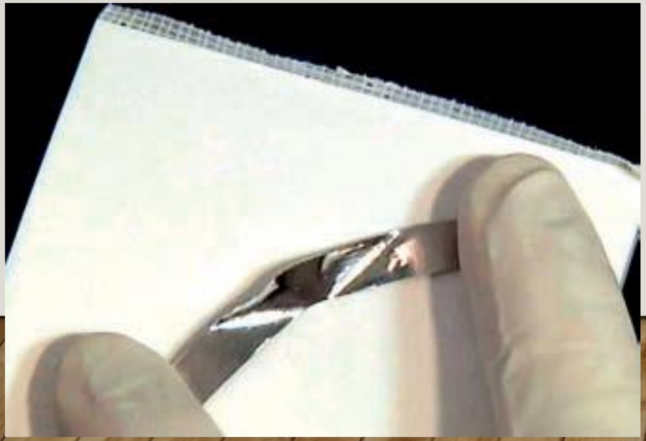
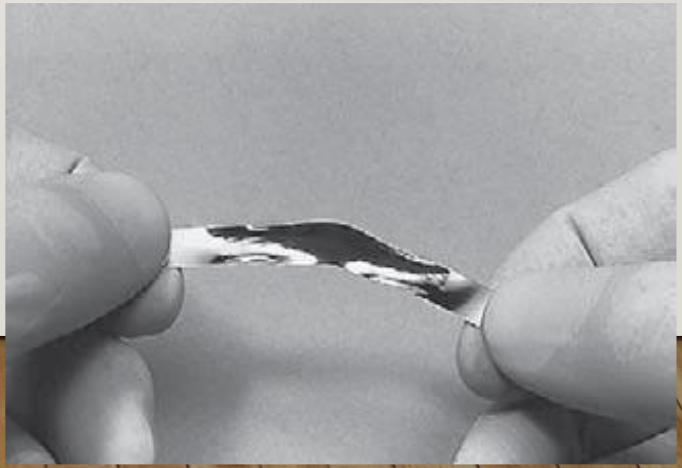
UNIVERSAL MATRIX

CONTOURING THE BAND(To achieve proper convex contour and proper adaptation)

DO IT ON PAPER PAD

USE EGG BURNISHER





POSITIONING OF THE UNIVERSAL MATRIX BAND IN THE RETAINER

PLACE THE GROOVE GINGIVALLY

LOOSEN THE SMALL KNARLED NUT

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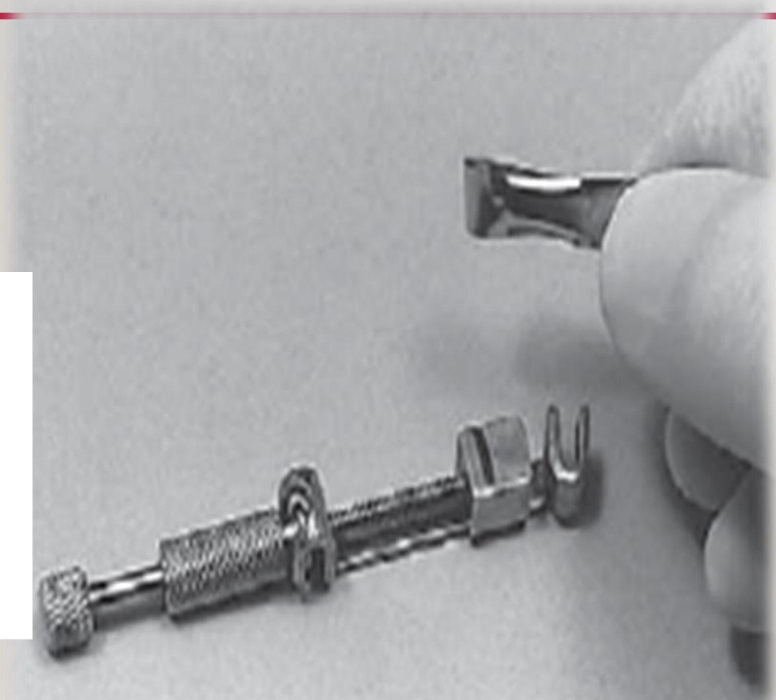
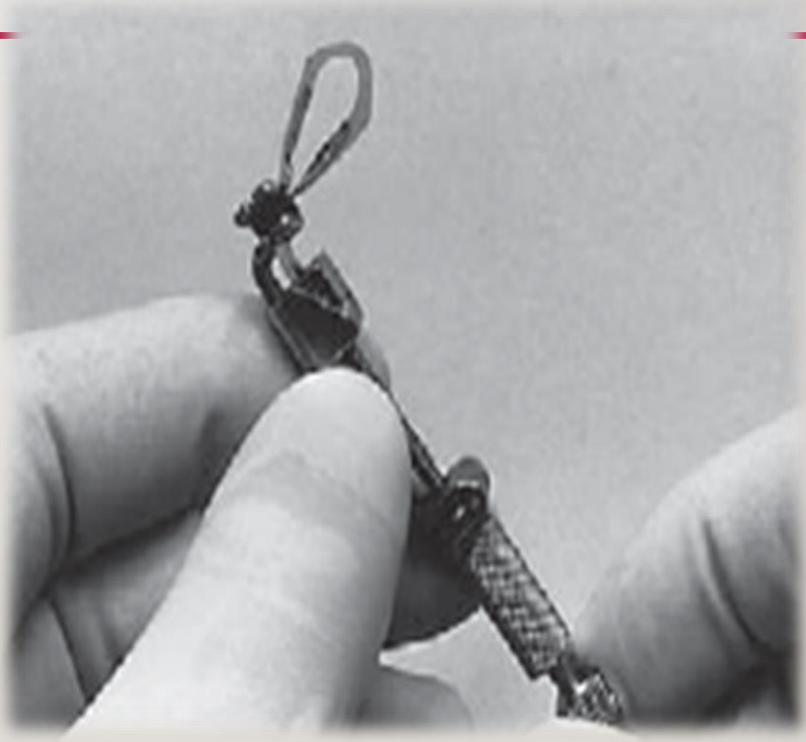
POSITIONING OF THE UNIVERSAL MATRIX BAND IN THE RETAINER



The convex surface (upward) & one concave (downward). The convexity of matrix band should put on the occlusal surface.

Form a loop



PASS IT IN THE GINGIVAL SL0T AND TIGHTEN THE LARGE KNARLED NUT.

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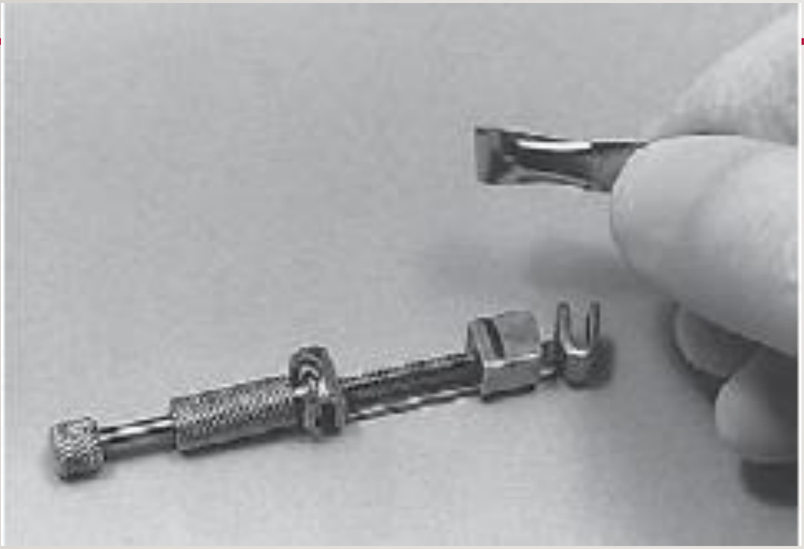
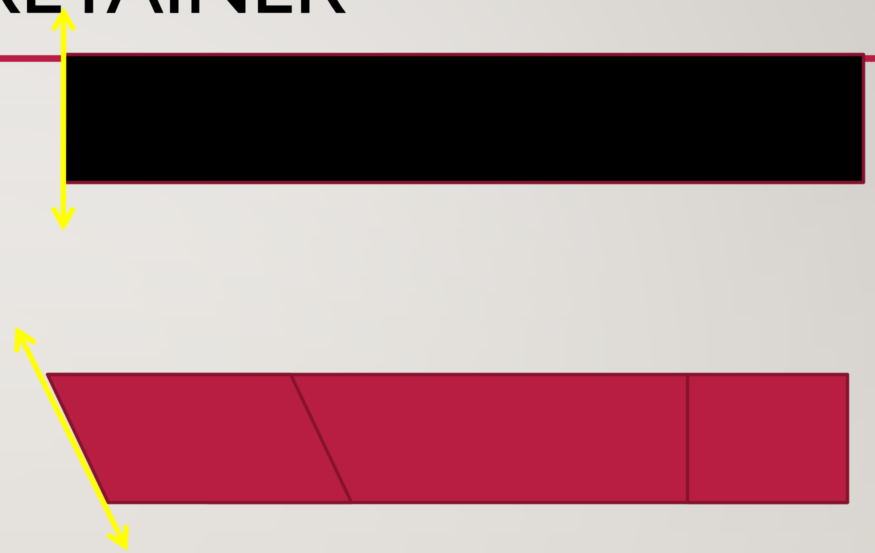
POSITIONING OF THE UNIVERSAL MATRIX BAND IN THE RETAINER



FORM LOOP WITH BAND AND POSITION



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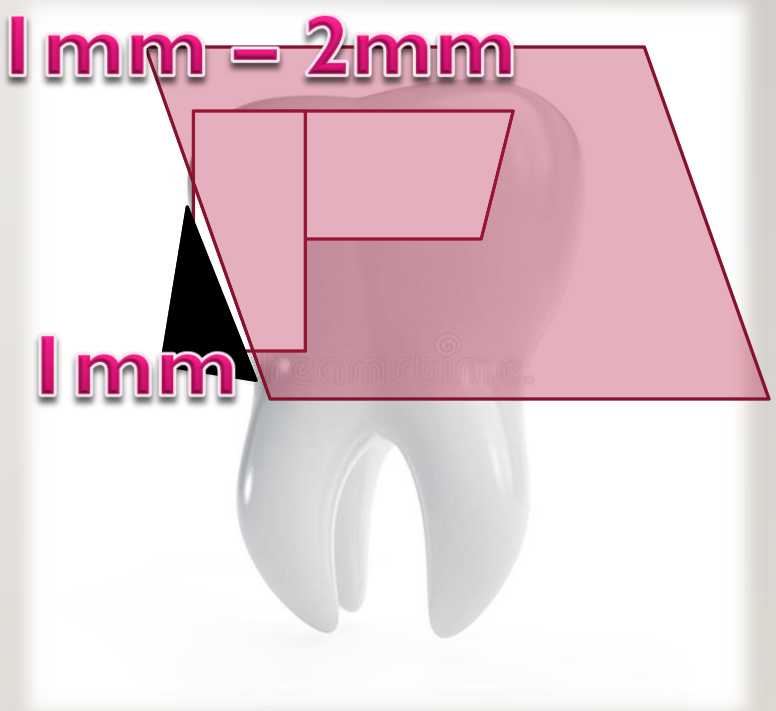
POSTIONING THE MATRIX RETAINER AND THE BAND OVER THE TOOTH

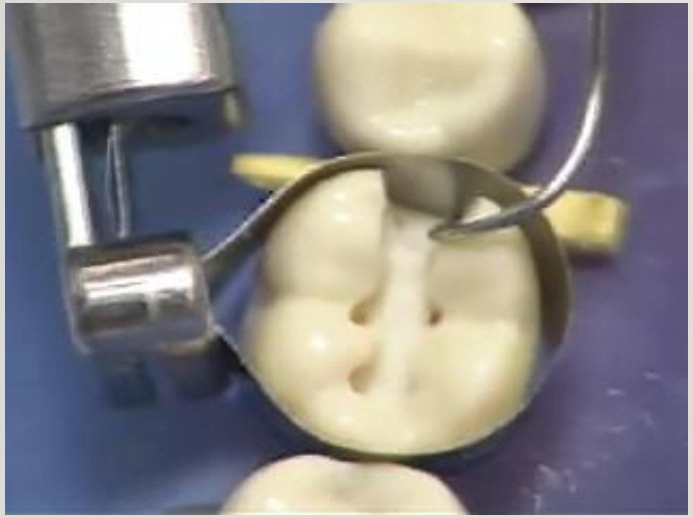


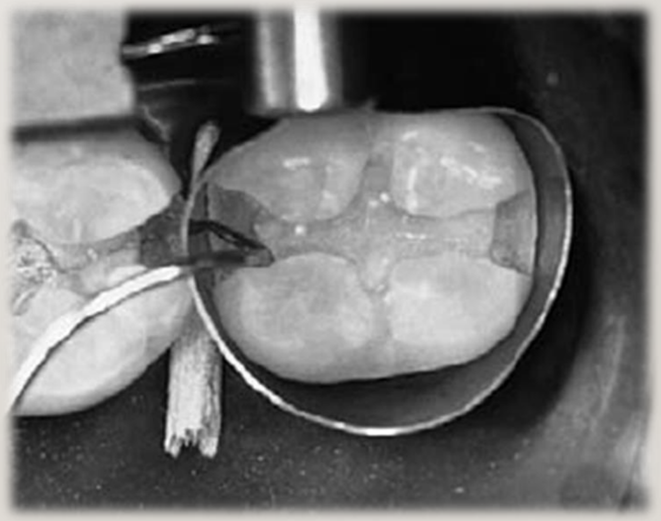
The matrix band should be 1 mm higher than the adjacent tooth marginal ridge and 1 mm below the gingival contact area

DO NOT TRAP THE RUBBER DAM IN THE BAND

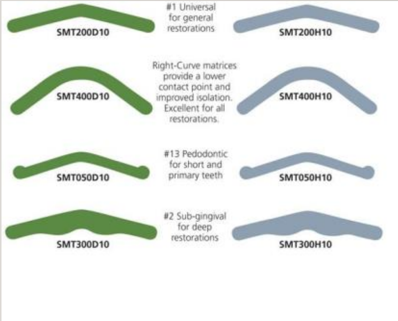
No gap between the band and the tooth

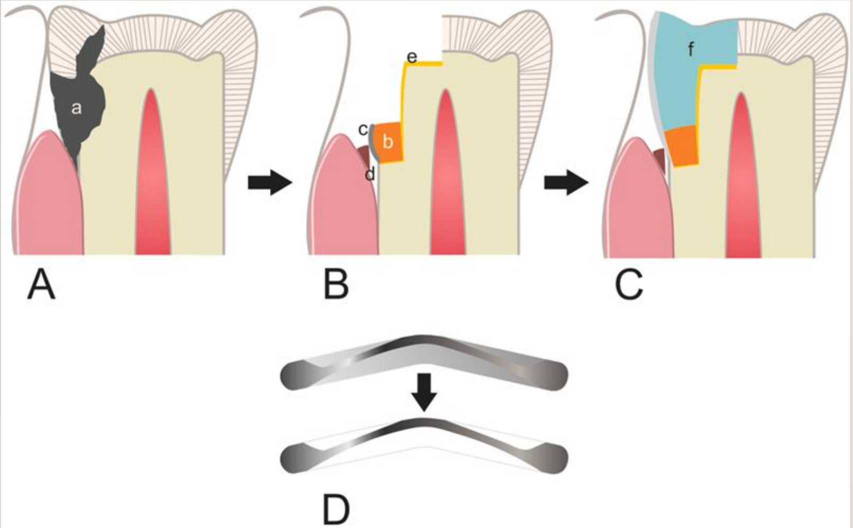
Use explorer tip (with pressure) between matrix band and gingival seat to ensure proper adaptation of the band to gingival margin.





IN CASE OF DEEP PROXIMAL MARGINS(SUBGINGIVAL RESTORATIONS) AND SHALLOW MARGINS





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USE THE MOUTH MIRROR FACIALLY AND LINGUALLY TO EVALUATE THE PROXIMAL CONTOUR

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PRE CONTOURED MATRIX (SECTIONAL MATRIX)

PALODENT

1. **The Sectional matrix is the best way to achieve strong contact point in a class II composite restorations.**

2. **Glass-fiber reinforced plastic tines maintain a fit on the tooth that complements the wedges and minimizes flash.**

3. Ring design promotes exceptional stability in the forceps during placement.The function of the ring is to adapt the sectional matrix in the cervical area.

4. V-shaped tines allow you to wedge from both sides before or after ring placement.

Sectional Matrix (PALODENT/TRIDENT systems)

AUTOMATRIX


INTRODUCTORY KIT
AutoMatrix®
Retainerless Matrix System
AutoMatrix

AUTOMATRIX

ADVANTAGE –

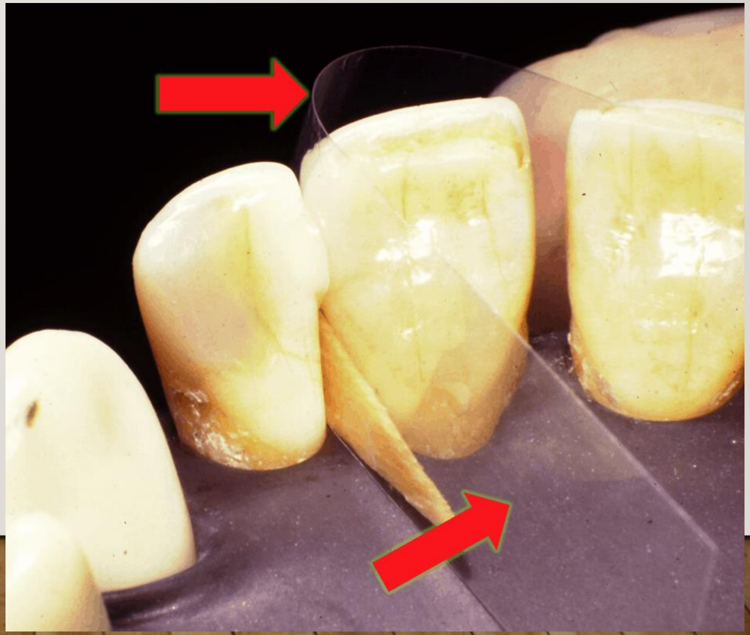
1. Autolock loop can be positioned on the facial and lingual surface with equal ease.
2. Improved visibility , because of absence of retainer

DIASDVANTAGE –

1. The bands are not precontoured
2. Development of physiological proximal contours is difficult.

POLYESTER / CELLULOID MATRIX SYSTEMS
Used in class III and class IV preparations

MYLAR MATRIX STRIPS



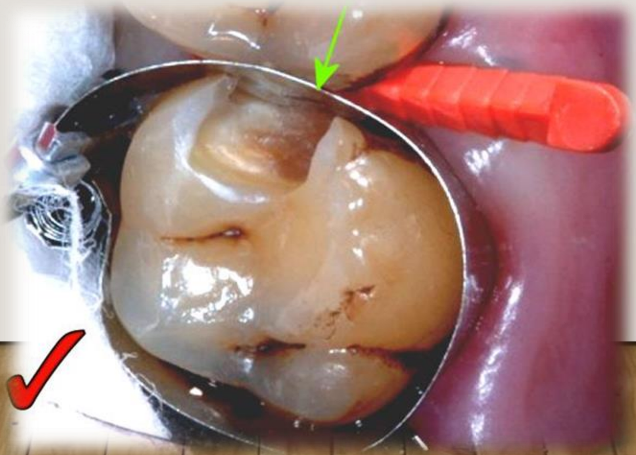
To restore Class-III and IV Cavity Preparation

WEDGES

Wedges are the devices that are placed to create rapid separation during tooth preparation and restoration and are placed in the gingival embrasure area

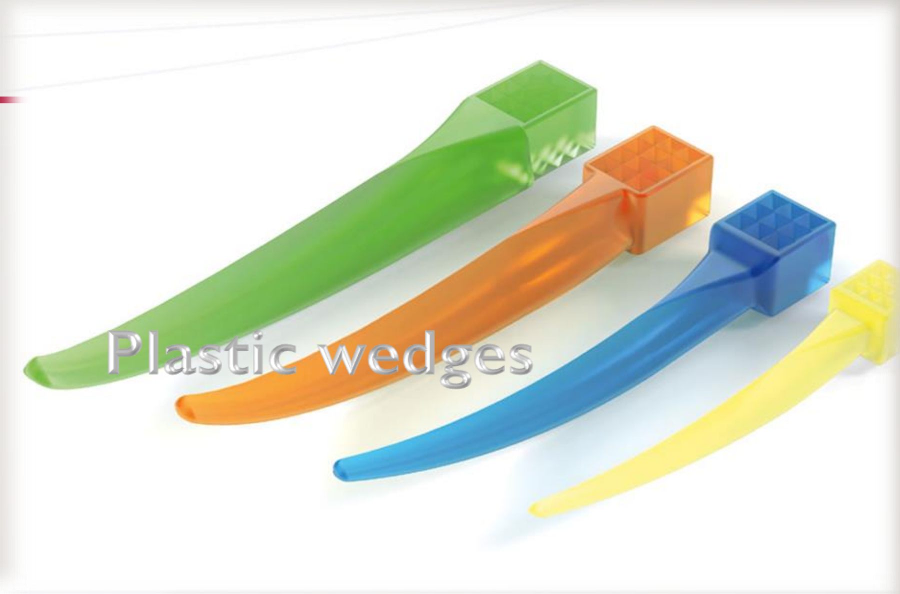
FUNCTIONS OF A WEDGE

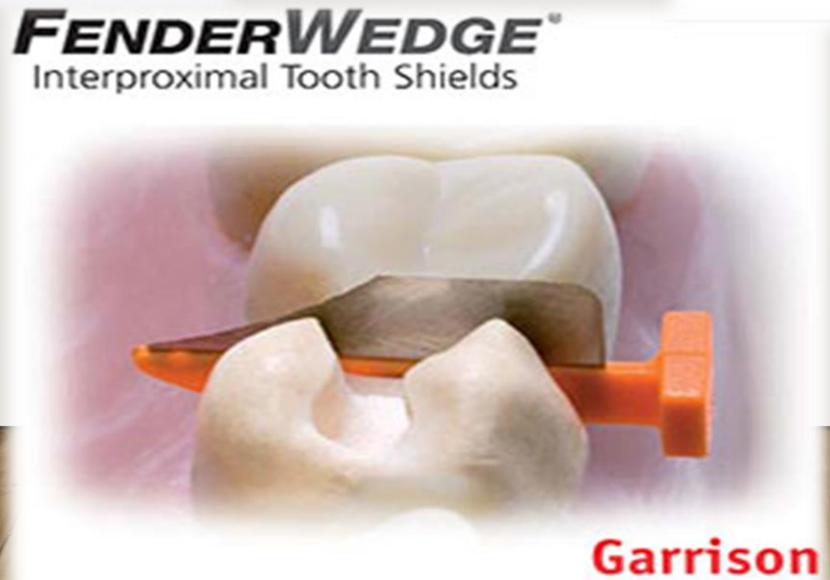
1. **Rapid separation of the tooth.**
2. **Protecting the underlying gingiva from unexpected trauma**
3. **To prevent gingival overhangs of restoration.**
4. **Compensating for the thickness of the matrix band to achieve proximal contact.**



For posterior restorations, the wedge is placed from the larger to the smaller interproximal gingival embrasure, typically from a lingual approach, just apical to the gingival margin.

TYPES OF WEDGES





CLASSIFICATION OF WEDGES

Based on the method of fabrication

Custom made

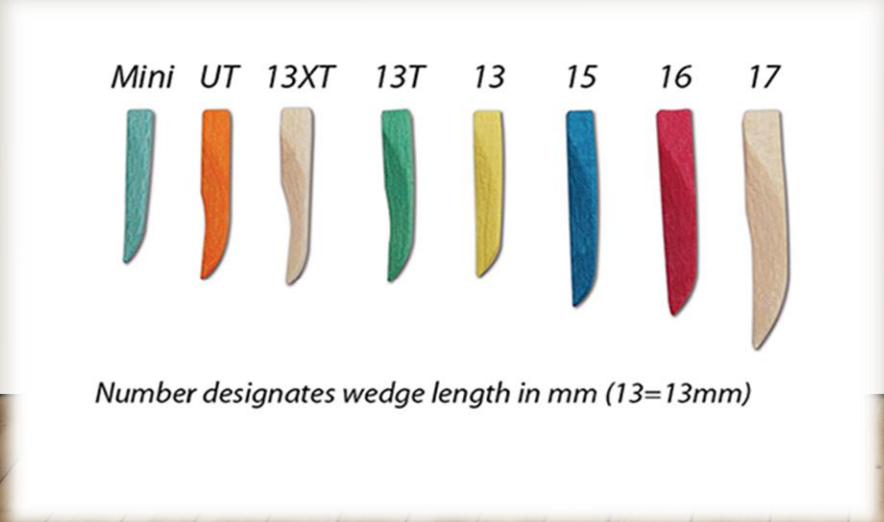
Prefabricated

Based on the material used

Wooden wedges

Resin /Plastic wedges

WODDEN WEDGES

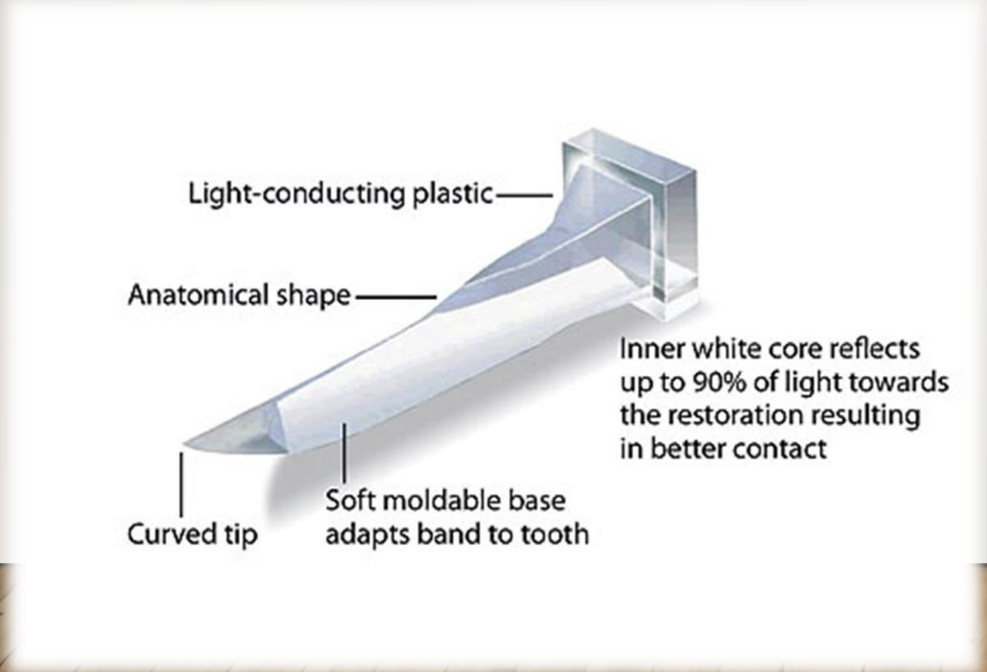


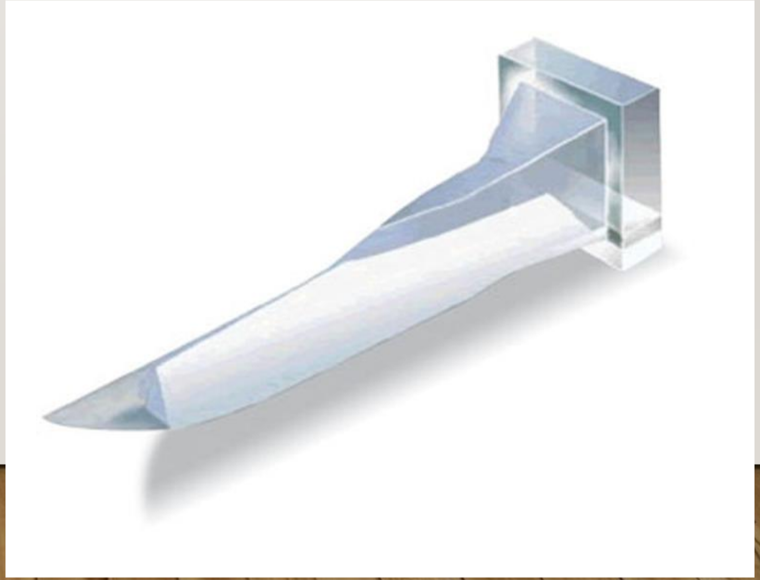
Advantage

- **Can be easily customized(Can be cut and trimmed by scalpel blade)**
- **They absorb water interiorly and swells up, which presses against the matrix to give more retention**

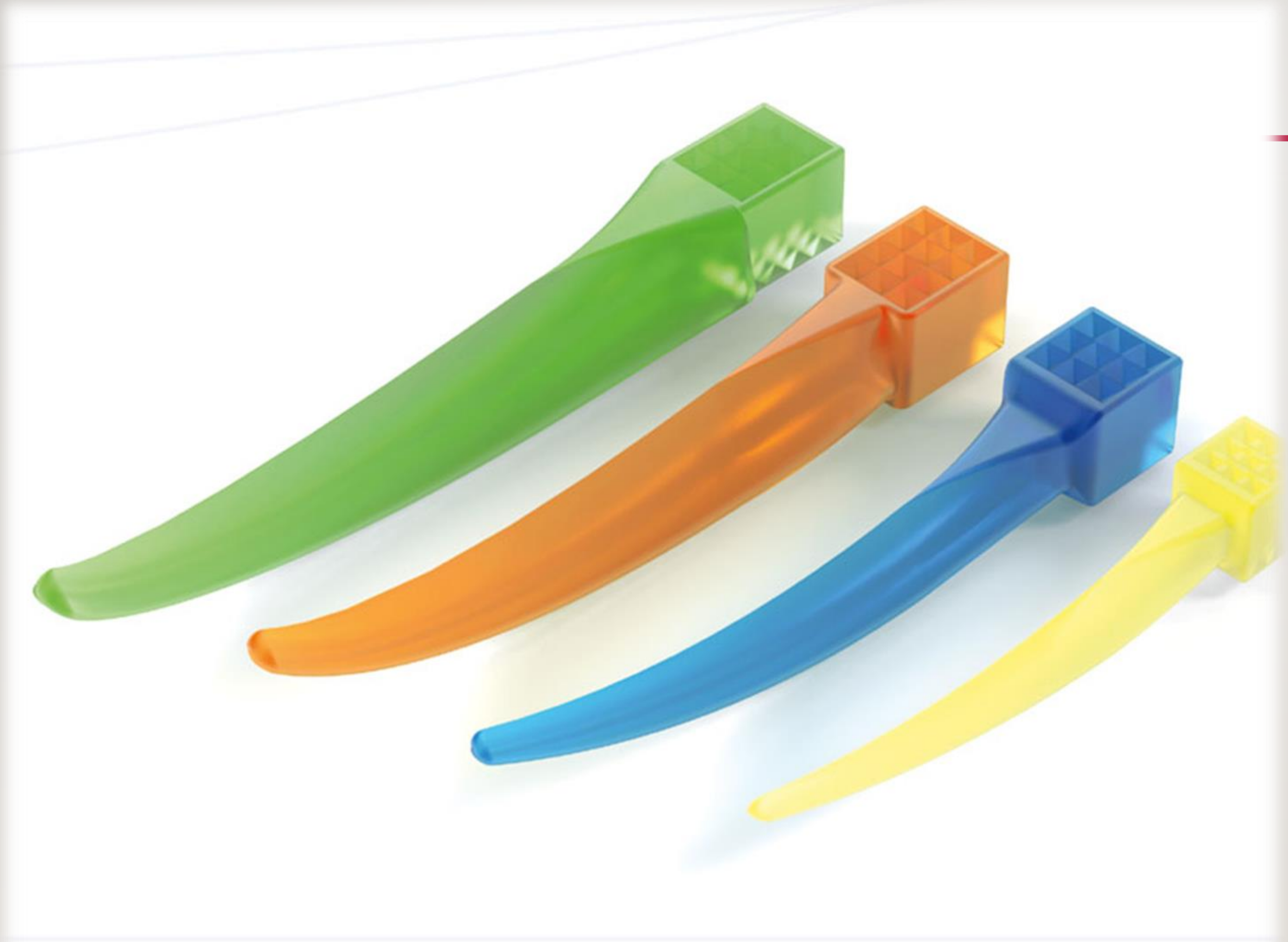
LIGHT TRANSMITTING WEDGES FOR COMPOSITE RESTORATIONS –

Transparent plastic wedges which are available with light reflecting property, helps to assist in directing the light into the interproximal areas during curing





PLASTIC WEDGES



Plastic wedges allows the visible light to pass through helpful in Composite restorations

PLASTIC WEDGES

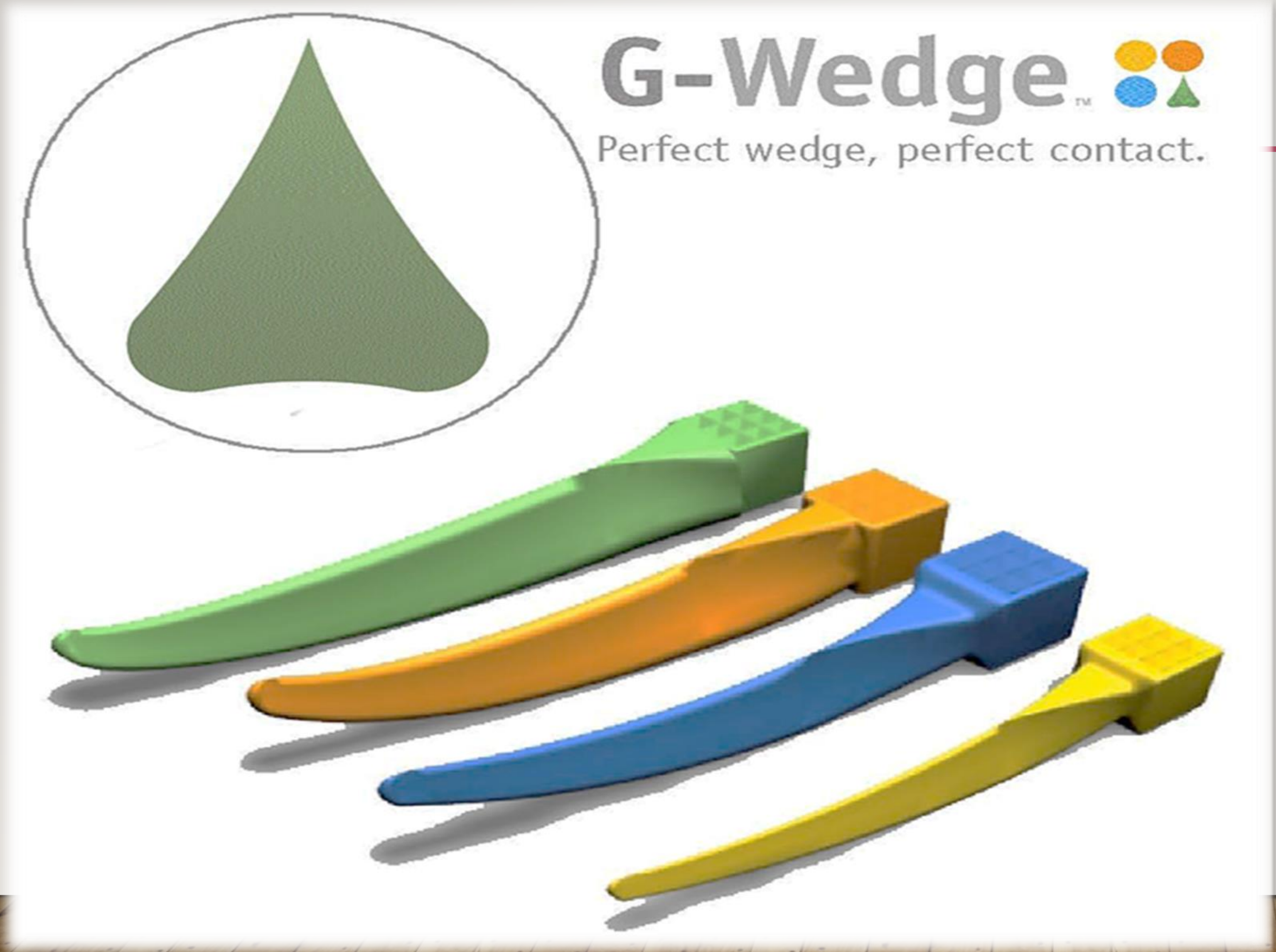
The Wand applicator provides the simplest placement technique ever.

No instrument such as cotton plier is required to place the wedge

Wedges with handles

WEDGE WANDS

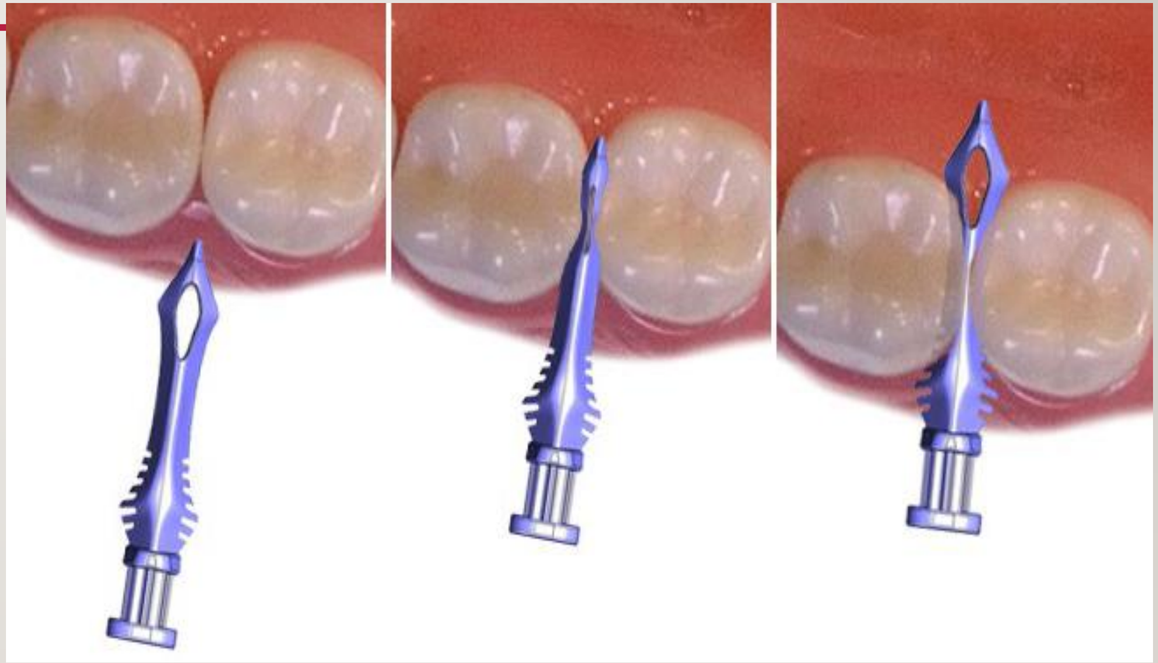
PLASTIC WEDGES



Traditional wedge placement with tweezer or cotton plier

G-Wedge™ gives you the same basic wedge shape as the **Wedge Wands®** without the applicator wand.

DIAMOND ANATOMICAL WEDGE



Diamond anatomical Wedge requires less effort to place than traditional wedges due to the diamond shaped cut-out in the tip, which collapses during insertion. The flexibility of the Diamond Wedge creates an excellent marginal adaptation.

FENDER MATRIX AND WEDGE

FenderWedge is a combination of a steel band and a plastic wedge.

Fender Wedge® will lend you a hand by "pre-separating" the teeth while you prepare the tooth.

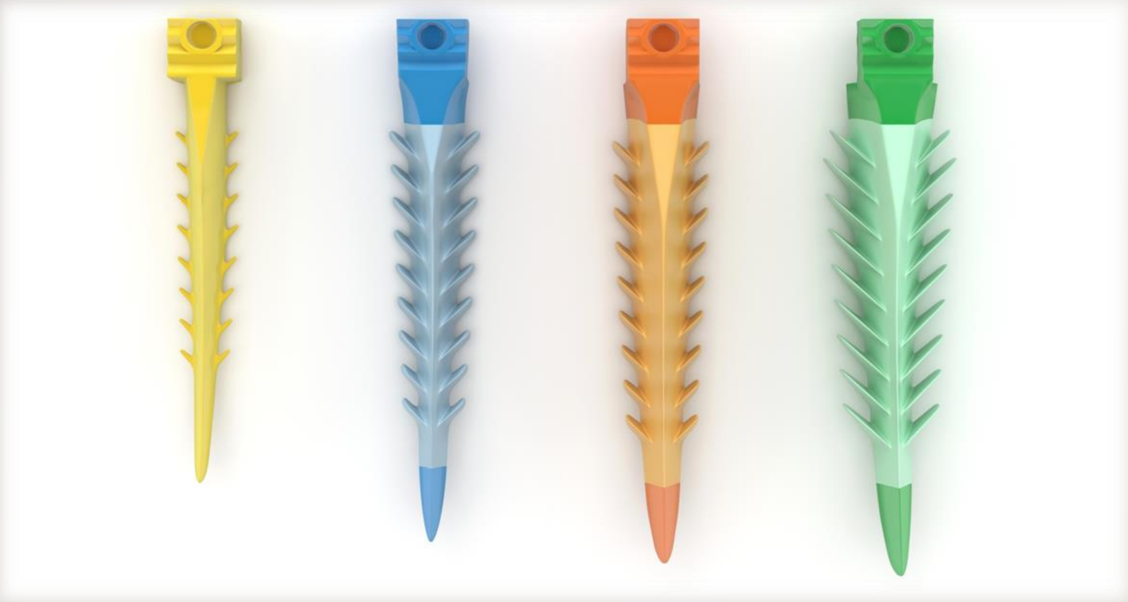
Protect the adjacent teeth


Large: 2.3mm
Medium: 1.9mm
Small: 1.4mm
X-Small: 1.0mm

As a added bonus, the firm stainless steel shield protects the adjacent tooth from inadvertent contact with the bur.

PLASTIC WEDGES

COMPOSI – TIGHT 3D
FUSION
ULTRA ADAPTIVE WEDGE



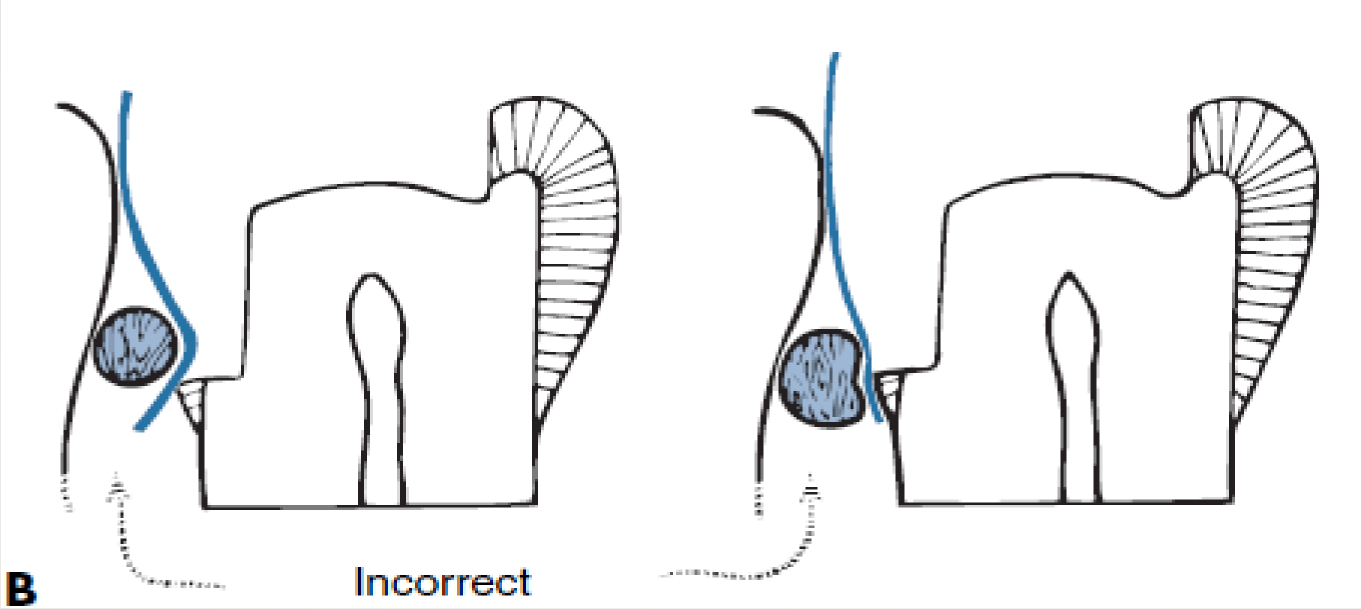
A soft, rubberized material is fused to the firm plastic core.

Advantages

It prevents backout of the wedge

The wedge gently molds itself to irregularities as you insert it providing an unsurpassed seal.

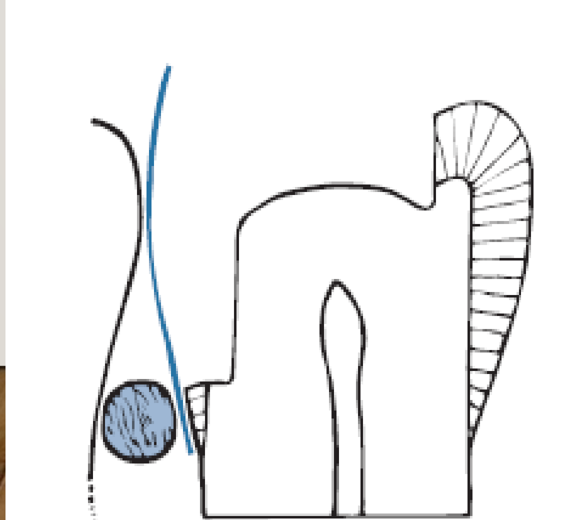
WEDGE PLACEMENT



If the wedge is placed occlusal to the gingival seat, the band will be pressed into the preparation, creating an abnormal concavity in the proximal surface of the restoration.

It should not interfere with the curvature of the matrix band

The wedge should be apical to the gingival seat that the band will be held tightly against the gingival margin.



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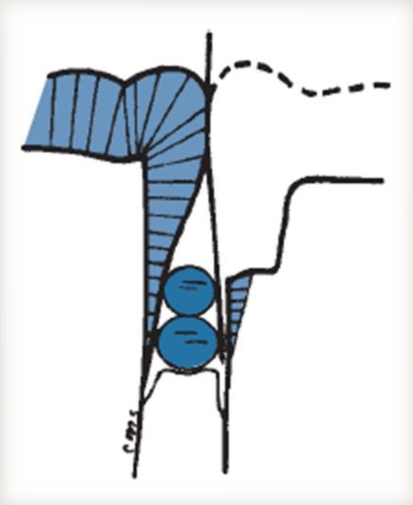
MODIFIED WEDGING TECHNIQUES

1.Piggy back wedging

In this technique, smaller wedge is placed over the first wedge, This method is useful when proximal box is shallow gingivally or inter-proximal gingival tissue level recession



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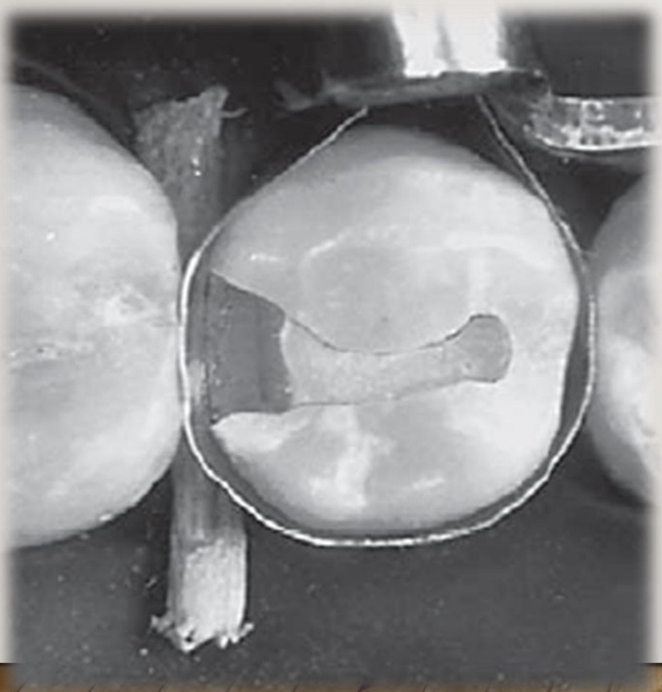
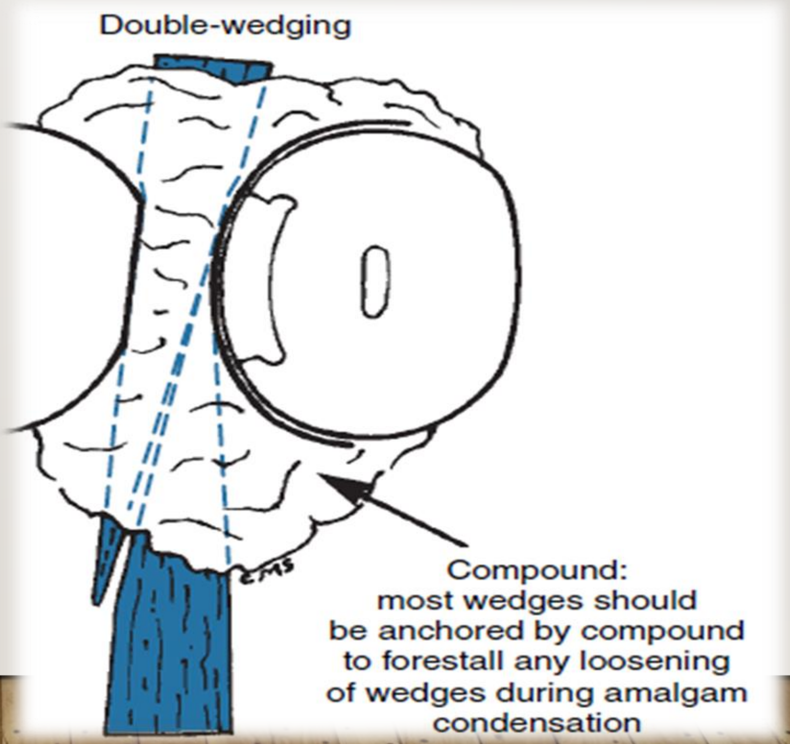
2.Double Wedging

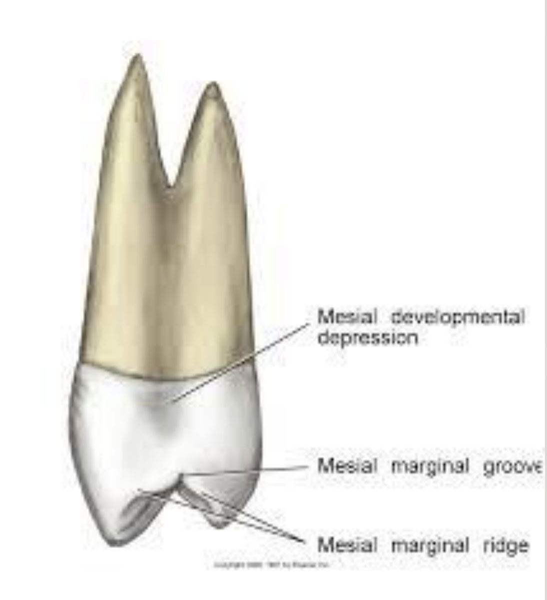
Double wedging ,as the name indicates one wedge from facial and other in lingual embraussre,may be used with buccolingually wide proximal boxes to provide maximal closure of band along gingival margin.

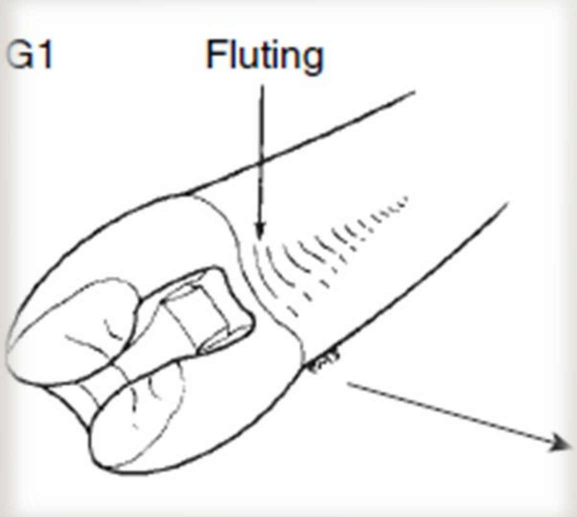


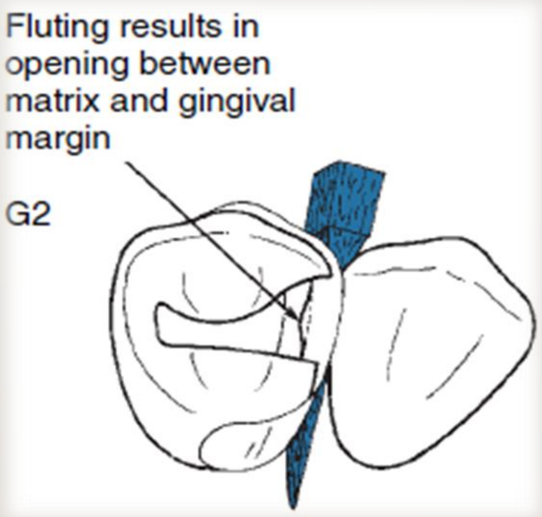


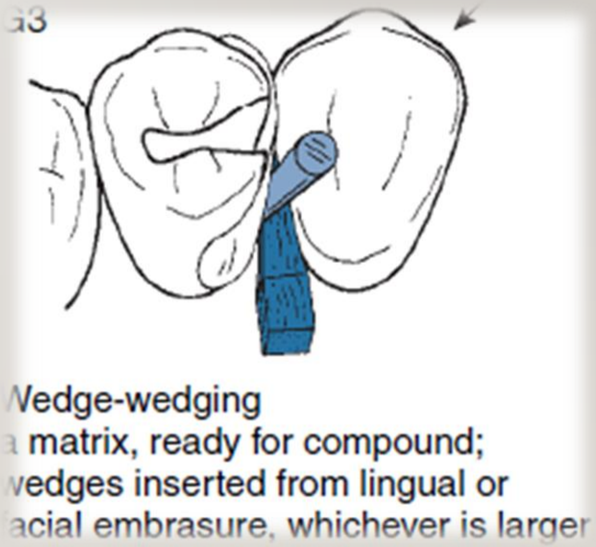
3. Wedge Wedging

Another technique may be used on the mesial aspect of maxillary first premolars to adapt the matrix to the fluted (i.e., Concave) area of the gingival margin ,*second wedge inserted between the first wedge and matrix band* .









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REMOVAL OF THE MATRIX AND WEDGE

First remove the retainer, then matrix band followed by wedge

DISPOSAL OF MATRIX BAND OR ANY SHARP ITEMS

Needles, scalpel blades, Local anesthetic cartridges, Burs, Matrix bands must be disposed of in the yellow sharps container

Reference:

1. Sturdevant's Art and Science of Operative Dentistry. 6th Edition
Chapter 14, Page 394-403

2. Sturdevant's Art and Science of Operative Dentistry.5th Edition
Chapter 17, Page 760-773

3. *https://www.garrisondental.com/en/wedges*

4. https://www.dentsplysirona.com/en-ca/products/restorative/accessories/matrix-systems.html